

PUBLIC DEMONSTRATION EDITION

Warehouse monitoring

Product description & operations playbook



Original AI-generated product illustration. Devices are conceptual, not manufactured products.

A practical guide to deploying and operating an ambient-condition and door-state monitoring kit in a small warehouse. Includes device selection, site planning, commissioning, alert response and maintenance.

DOCUMENT CONTROL. Release details

| Field        | Value                                            |
|--------------|--------------------------------------------------|
| Document ID  | AW-PB-001                                        |
| Edition      | 1.0 / 09 September 2026                          |
| Audience     | Warehouse supervisor, installer, support analyst |
| Example site | WH-01: dry storage, packing and receiving        |

Synthetic document. AisleWatch, its devices, site, specifications, procedures and records are fictional. Created for public blog, GitHub and retrieval demonstrations. All photographs are AI-generated; diagrams and tables are original. No customer records or third-party source images are included.

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Reading convention: model names (such as TH-20) identify product types; device IDs (such as TH-03) identify installed units. Table and figure numbers belong to their parent section.

## 01

# Product overview

## 01.1 Purpose and operating boundary

AisleWatch helps a warehouse team notice sustained changes in ambient temperature, relative humidity and door state. A gateway collects sensor readings and sends them to a browser console. Staff investigate alerts and record a resolution. The kit provides visibility; it does not control building equipment.

The reference deployment, WH-01, contains three work zones and six connected devices: one gateway, three ambient sensors and two door contacts. The supervisor owns the zone profiles and signs off acceptance before a shift relies on the console.

T01-1. Supported scope for this fictional release

| Use case           | Product behavior                                 | Boundary                                   |
|--------------------|--------------------------------------------------|--------------------------------------------|
| Ambient monitoring | Shows temperature and relative humidity per zone | Does not measure the inside of packages    |
| Door awareness     | Tracks open/closed transitions and open duration | Does not lock doors or detect people       |
| Exception handling | Creates alerts after configured persistence      | Acknowledge does not resolve the condition |
| Incident review    | Exports readings and operator events             | No camera feed or personnel identification |

## 01.2 Roles and ownership

Installer: mounts devices, maps their IDs to zones and completes the acceptance checklist. Shift operator: reviews active alerts and records observations. Supervisor: approves configuration changes and unresolved handovers. Support analyst: investigates device health and export anomalies.

## 01.3 Reference outcome

A receiving door left open can be identified through a door-duration alert. If the adjacent ambient sensor also exceeds its profile, staff can connect the two observations during investigation. The console reports correlation in time; it does not prove why temperature changed.

The example is intended for ordinary dry-goods operations. No food-storage, pharmaceutical, life-safety or regulatory suitability is claimed. All operating limits in this playbook are invented demonstration values.

## 02

## Kit and device identification

The WH-01 starter kit uses three device types. Register the printed device ID during commissioning; appearance identifies the type, but cannot identify an individual installed unit.



F02-1. Conceptual kit photograph. Relative sizes are illustrative; identify shapes visually and confirm the model against T02-1.

T02-1. WH-01 kit inventory

| Model / item         | Qty   | Recognition and role                                                                                   |
|----------------------|-------|--------------------------------------------------------------------------------------------------------|
| GW-100 gateway       | 1     | Charcoal enclosure with two antennas; collects local device traffic. Installed ID: GW-01.              |
| TH-20 ambient sensor | 3     | White enclosure with lower ventilation grille; measures temperature and humidity. IDs: TH-01 to TH-03. |
| DC-10 door contact   | 2     | Two-piece white body and separate narrow magnet; reports door state. IDs: DC-01 and DC-02.             |
| Mounting pack        | 1     | Brackets, alignment labels and device-ID labels; not shown.                                            |
| Gateway accessories  | 1 set | 12 V DC adapter and Ethernet lead; not shown.                                                          |

### 02.1 Receiving check

Match quantities to T02-1. Check housings, grilles and mounting brackets for damage. Keep each door-contact body with its magnet. Record a missing accessory before installation; do not substitute an unapproved gateway power source.

## 03

## Technical specifications

The following values define the fictional AW-PB-001 demonstration model. They are deliberately explicit so configuration and troubleshooting examples can be checked against a single specification table.

T03-1. Device operating specifications

| Parameter         | GW-100                       | TH-20                               | DC-10                              |
|-------------------|------------------------------|-------------------------------------|------------------------------------|
| Primary input     | 12 V DC adapter              | 2 × AA cells                        | 1 × CR2032 cell                    |
| Operating ambient | 0 to 45 °C                   | 0 to 50 °C                          | 0 to 50 °C                         |
| Humidity exposure | 10-85% RH;<br>non-condensing | 10-90% RH;<br>non-condensing        | 10-90% RH;<br>non-condensing       |
| Measurement       | Device health                | Temperature: 0-50 °C; RH:<br>10-90% | Open / closed                      |
| Nominal accuracy  | Not applicable               | ±0.5 °C; ±3% RH                     | Not applicable                     |
| Reporting         | 60 s heartbeat               | 60 s sample and report              | On state change; 60 s<br>heartbeat |
| Local retention   | 10,000 queued records        | No historical backfill              | No historical backfill             |
| Capacity / limit  | Up to 32 paired sensors      | Grille must remain<br>exposed       | Closed gap: 3-8 mm<br>inclusive    |
| Placement         | Indoor; Ethernet required    | Indoor; shaded airflow              | Indoor; aligned marks              |

### 03.1 Capacity planning

WH-01 uses five of the gateway's 32 sensor slots. The gateway itself is not counted as a paired sensor. Additional gateways are separate installations; this edition does not define roaming between gateways.

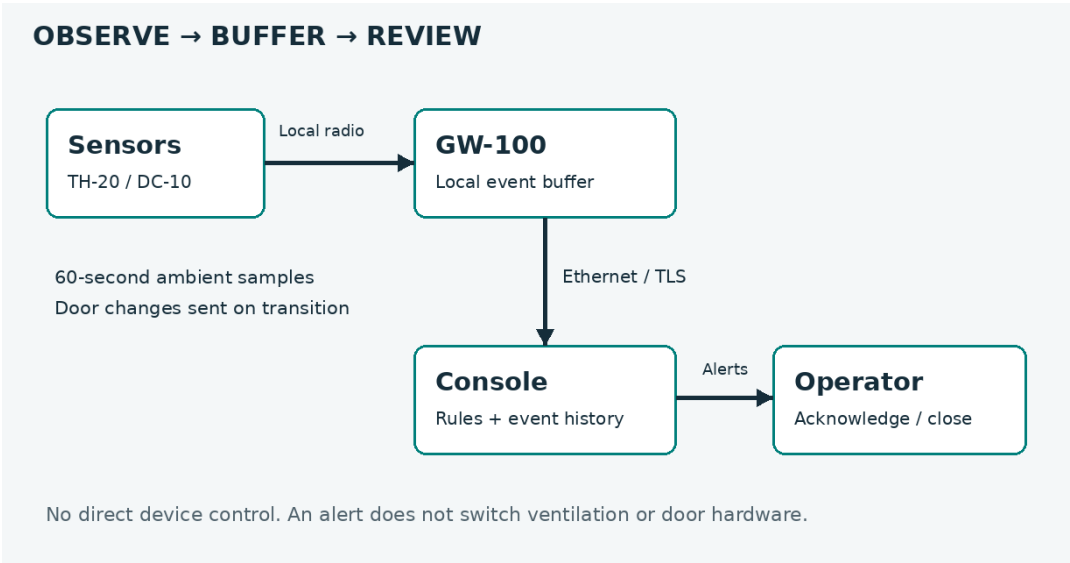
### 03.2 Reading limits

A value outside the measurement range is flagged as invalid instead of being clipped to the nearest boundary. Invalid readings do not advance alert persistence or clear an existing condition. A sensor can be online while its latest measurement is invalid.

Battery life and radio range are not specified in this edition. Do not infer either from the reporting interval or from the illustrative floor plan.

04

System architecture and buffering



F04-1. Logical event path. Local radio links join sensors to the gateway; Ethernet connects the gateway to the console service.

T04-1. Behavior during connectivity changes

| State                | Expected behavior                                                                      | Operator action                                    |
|----------------------|----------------------------------------------------------------------------------------|----------------------------------------------------|
| Normal uplink        | New records reach the console; server rules evaluate current events.                   | Review age and condition together.                 |
| Internet unavailable | Gateway queues up to 10,000 records. Cloud alert evaluation cannot see new conditions. | Use local checks; mark the console as stale.       |
| Uplink restored      | Oldest queued records upload first. Duplicate event IDs are ignored.                   | Confirm sync time and inspect the incident window. |
| Queue full           | Oldest queued records are overwritten. Gateway reports a dropped-record count.         | Document the gap; do not claim complete history.   |
| Sensor radio lost    | Gateway has no missing sensor samples to backfill.                                     | Investigate placement, battery and device health.  |

04.1 Event-time semantics

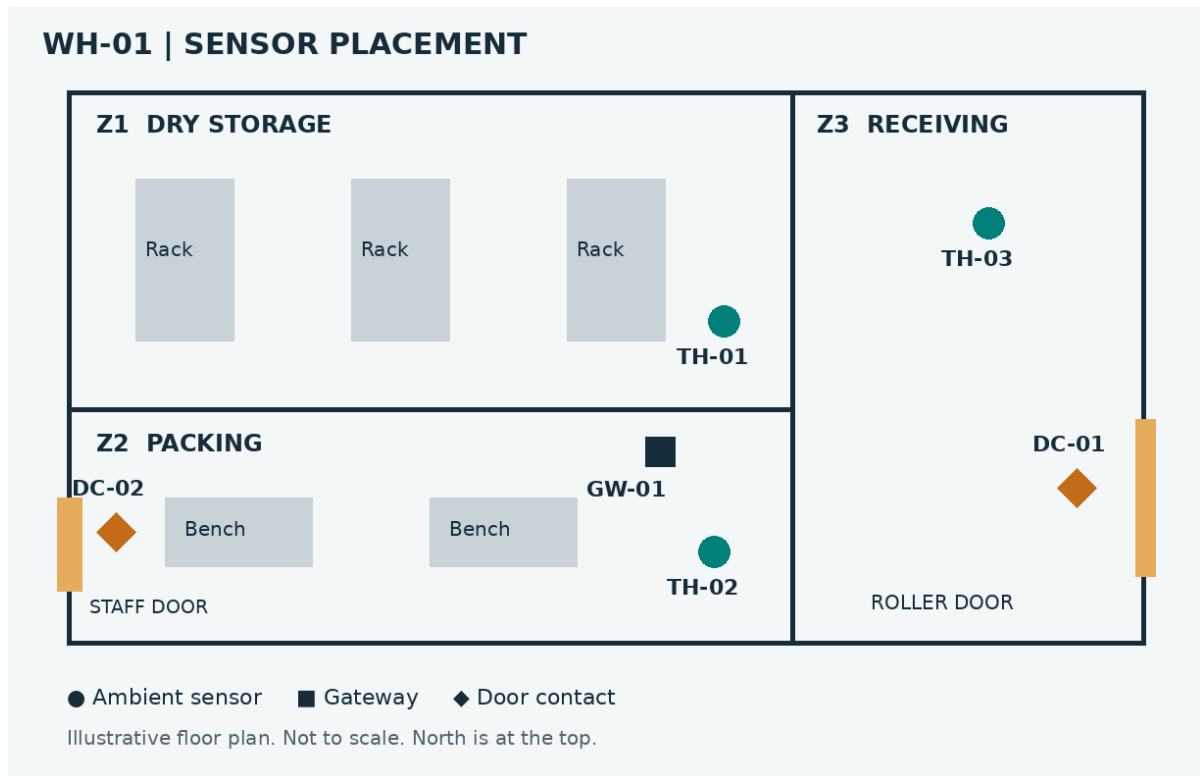
Each event has an observation time and a receipt time. Delayed records update history using observation time. Records more than 120 seconds old at receipt do not open new live environmental alerts; they remain available for incident review.

The gateway has no backup battery in this edition. Loss of gateway power stops collection; the local queue only helps while the gateway remains powered.

## 05

## Warehouse placement plan

Use F05-1 to map installed device IDs to physical zones and entrances. The plan is schematic: it supports location lookup, not distance measurement or proof of radio coverage.



F05-1. WH-01 reference floor plan and installed device locations. Device-to-zone and device-to-door relationships are shown in the image.

T05-1. Approved WH-01 zone profiles

| Zone | Purpose     | Profile | Temperature / humidity ceiling |
|------|-------------|---------|--------------------------------|
| Z1   | Dry storage | P-STOCK | 28 °C / 65% RH                 |
| Z2   | Packing     | P-WORK  | 30 °C / 70% RH                 |
| Z3   | Receiving   | P-DOCK  | 32 °C / 75% RH                 |

### 05.1 Placement review

Walk the site with the installer and compare each physical ID to the plan. Test both doors while watching the matching device record. Apply the profile belonging to the sensor's mapped zone, then use the persistence rules in Section 08.

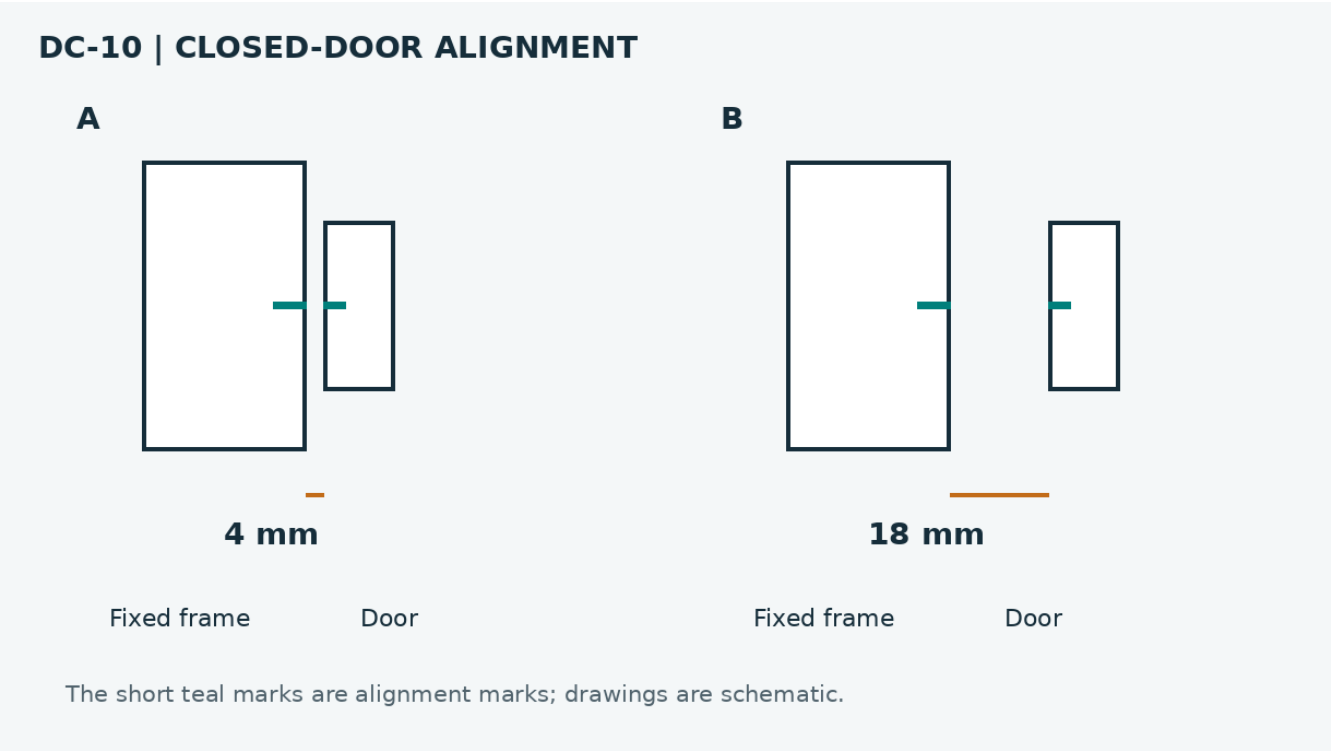
Relocation requires a new location check and profile review. Moving a device without updating its mapping can produce a technically valid alert against the wrong zone limit.

06

Installation standards

T06-1. Mounting requirements

| Device | Required placement                                                        | Reject the installation if                                                     |
|--------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| GW-100 | Stable shelf, 1.5-2.0 m above floor; both antennas upright.               | Enclosed in a metal cabinet or powered by an undocumented adapter.             |
| TH-20  | 1.5 m above floor; at least 1 m from a door, heater or supply-air outlet. | Grille is obstructed, housing receives direct sunlight, or moisture condenses. |
| DC-10  | Body on fixed frame; magnet on moving door; alignment marks opposite.     | Closed-door gap is outside 3-8 mm or parts are vertically misaligned.          |



F06-1. Two closed-door installation examples. Compare each illustrated gap with the DC-10 requirement in T06-1.

06.1 Installation sequence

1. Confirm the zone and physical mounting point. 2. Secure the bracket without covering ventilation. 3. Attach the device and apply its ID label. 4. Measure the door-contact gap when relevant. 5. Photograph the completed installation and run Section 07 checks.

Why this order: establishing location before pairing reduces mapping mistakes; testing after final mounting catches alignment or radio problems that bench testing misses.

## 07

## Commissioning and acceptance

Create site WH-01, enroll GW-01 and pair the five sensors. Use the device map in Section 05, then enter the zone profiles before testing alerts. Commissioning is complete only when the supervisor accepts every required check.

T07-1. Acceptance record template

| Check                | Method                                                                | Pass criterion                                                               |
|----------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------|
| Identity and mapping | Compare each installed ID with F05-1; open each door once.            | All five sensor records match their actual locations.                        |
| Freshness            | Observe three consecutive ambient reports.                            | Each arrives no more than 120 s after observation.                           |
| Ambient plausibility | Compare with a reference meter after 15 min at the same location.     | Difference no greater than 1.0 °C and 5% RH.                                 |
| Door operation       | Start closed; open and close the door three times.                    | All six transitions are visible; final state is closed.                      |
| Rule handling        | Use simulator to send six 60 s-spaced samples above the zone ceiling. | High-temperature alert opens on sample six, after 300 s.                     |
| Acknowledgement      | Operator acknowledges the test alert with a note.                     | Actor, time and note appear; condition is still active.                      |
| Recovery             | Send three 60 s-spaced samples below the recovery limit.              | Condition recovers after 120 s; closure is recorded separately.              |
| Uplink interruption  | Disconnect Ethernet for 3 min while gateway remains powered.          | Queued records appear after reconnect; no new live alert from stale records. |

### 07.1 Approval record

Record installer, supervisor, completion time, device IDs and any deviation in the commissioning ticket. Re-run a failed check after correcting the cause. A signed exception must describe the limitation and who will monitor it.

### 07.2 Simulator boundary

The simulator labels all events as test data and keeps them separate from live operations. Use simulated threshold crossings to verify rules; do not heat a warehouse area merely to trigger an acceptance alert.

## 08

## Alert rules and escalation

Use the ceiling from T05-1 for the ambient sensor's zone. A breach requires a value strictly greater than that ceiling. Start the timer on the first qualifying sample; reset it on a non-qualifying sample or a gap greater than 90 seconds.

T08-1. Rule and recovery definitions

| Rule               | Open condition                                         | Automatic recovery                                        |
|--------------------|--------------------------------------------------------|-----------------------------------------------------------|
| High temperature   | Above zone ceiling continuously for 300 s.             | At or below ceiling minus 1 °C for 120 s.                 |
| High humidity      | Above zone ceiling continuously for 600 s.             | At or below ceiling minus 3 percentage points for 120 s.  |
| Door open too long | Door remains open for more than 120 s.                 | Closed transition received.                               |
| Sensor offline     | No event or heartbeat for more than 300 s.             | First new valid heartbeat or event.                       |
| Low battery        | Battery estimate below 20% in two consecutive reports. | Battery estimate at least 25% in two consecutive reports. |

### 08.1 Timing example

For P-STOCK, readings above 28 °C at 10:00, 10:01, 10:02, 10:03, 10:04 and 10:05 open the temperature alert at 10:05. A reading of exactly 28 °C breaks the opening sequence. After an alert opens, readings at or below 27 °C at 10:10, 10:11 and 10:12 recover the condition at 10:12.

### 08.2 Responsibility and escalation

The shift operator owns all new alerts and acknowledges within 5 minutes. An unacknowledged alert goes to the supervisor after 5 minutes. If the condition remains active 15 minutes after opening, the supervisor arranges an on-site check and records the next review time. These are example operating targets, not a service guarantee.

### 08.3 Acknowledge, recover, close

Acknowledge records that a person has accepted the task. Recover means the measured condition meets the rule's recovery test. Close is the operator's final action after recovery and a resolution note. Acknowledgement never suppresses a continuing condition.

09

# Daily operations

Begin with data freshness, then inspect active alerts. A displayed reading is useful only when its age is understood. Keep acknowledged alerts in view until the condition recovers and the operator closes the record.

AISLEWATCH / WH-0114:10 UTC

Devices online6 / 6

Open alerts2

Gateway sync14:10

ACTIVE ALERTS

| Device | Rule               | Current | Age   | Status |
|--------|--------------------|---------|-------|--------|
| TH-03  | High temperature   | 31.8 °C | 7 min | Open   |
| DC-01  | Door open too long | Open    | 4 min | Ack.   |

Filters: Current alerts | All zones

Snapshot: 09 Sep 2026. Values are synthetic.

F09-1. Synthetic console snapshot. Current readings can differ from the values that originally opened an alert. "Ack." means acknowledged, not resolved.

T09-1. Shift routine

| When            | Action                                                         | Evidence to record                                 |
|-----------------|----------------------------------------------------------------|----------------------------------------------------|
| Start of shift  | Check online count, gateway sync time and open alerts.         | Operator, time and unresolved incident IDs.        |
| During an alert | Confirm device, mapped zone, current value and recent history. | Observed condition, action taken and next review.  |
| At recovery     | Confirm the recovery rule is satisfied; close with a note.     | Cause if known; otherwise state it is unconfirmed. |
| Shift handover  | Review active and acknowledged alerts with the next operator.  | Owner and next check time for each open incident.  |

## 09.1 Interpreting a remaining alert

A temperature alert may remain active even when its current reading is below the opening ceiling. Check the lower recovery limit and the recovery timer in Section 08 before treating the display as inconsistent.

## 10

## Troubleshooting and inspection



F10-1. Inspection case INS-014: TH-20 at a mounting point. Identify the visible condition and select the matching action from T10-1.

### T10-1. Symptom-to-action guide

| Code / symptom                             | Check                                                              | Corrective action                                                                 |
|--------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| AIR-01: Slow ambient response              | Look for tape, labels, dust or packaging over the grille.          | Remove the obstruction; allow 15 min to settle; repeat the reference-meter check. |
| DOOR-02: Closed door reads open            | Measure closed gap and inspect alignment marks.                    | Restore 3-8 mm gap and alignment; repeat three open/close cycles.                 |
| LINK-03: Multiple sensors stale            | Check gateway power and Ethernet before replacing batteries.       | Restore gateway connectivity; review history for gaps.                            |
| PWR-04: One sensor offline                 | Inspect battery seating, battery level and local mounting changes. | Fit specified battery if needed; confirm new events and mapping.                  |
| RULE-05: Alert stays after acknowledgement | Check active condition and recovery timer.                         | Resolve condition; wait for recovery; close with a note.                          |

### 10.1 Escalation package

If the fault persists, attach the device ID, site, first observed time, last event time, a mounting photograph and checks already performed. Do not infer physical damage solely from a missing reading.

## 11

# Maintenance and replacement

Maintenance preserves the connection between a physical device and the record people rely on. Log every battery replacement, mounting change and profile adjustment against the installed device ID.

T11-1. Maintenance schedule

| Frequency / trigger       | Task                                                                           | Completion evidence                                       |
|---------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------|
| Weekly                    | Inspect ambient grilles, brackets and door-contact alignment.                  | Inspection date, device ID and photograph for any defect. |
| Monthly                   | Repeat one reference-meter comparison per zone; review dropped-record counts.  | Comparison readings and any acknowledged data gaps.       |
| Battery alert             | Replace with the model-specific battery in T03-1; check seating and polarity.  | Battery replacement event and two healthy reports.        |
| After relocation          | Update F05-1-equivalent site mapping and reapply the destination zone profile. | New location, profile and repeated acceptance checks.     |
| After gateway replacement | Enroll new gateway and pair the retained sensors; validate history boundaries. | New gateway ID, paired device list and cutover time.      |
| Quarterly                 | Review operator access and unresolved incident records.                        | Supervisor approval and access changes logged.            |

## 11.1 Replacement without identity confusion

Retire the old device record rather than reusing its identifier for a different unit. Give the replacement a new device ID and link both records in the maintenance ticket. Historical events remain associated with the original device.

## 11.2 Cleaning and return to service

Use a dry soft brush around the ambient grille and a dry cloth on the housing. Do not spray liquid into openings. After cleaning, allow 15 minutes for readings to settle, then compare against a reference meter using T07-1.

## 11.3 Change control

A profile change applies to new evaluations from its effective time. Preserve the previous threshold in each existing alert record. Record why a limit changed so an incident review can reproduce the rule that was active.

## 12

## Data export and document provenance

### 12.1 Export contract

CSV exports use UTF-8, one event per row and UTC timestamps. Empty metric cells mean not applicable or unavailable; they never mean zero. Filter exports by site and observation-time range, and retain receipt time to identify delayed uploads.

T12-1. Core event fields

| Field                        | Meaning / example                                            |
|------------------------------|--------------------------------------------------------------|
| event_id                     | Unique event key; used to remove duplicate uploads.          |
| site_id / device_id          | Installed site and device, e.g. WH-01 / TH-03.               |
| model / zone_id              | Device type and zone mapping effective when observed.        |
| observed_at / received_at    | ISO 8601 UTC timestamps with Z suffix.                       |
| event_type                   | ambient_reading, door_state, heartbeat or operator_action.   |
| temperature_c / humidity_pct | Numeric ambient values; empty for door-only events.          |
| door_state / battery_pct     | open or closed when applicable; estimated charge percentage. |
| quality / is_test            | valid or invalid measurement; true for simulator events.     |
| profile_id / profile_version | Rule configuration used for the event evaluation.            |

### 12.2 Public provenance

This document was created as an original fictional demonstration corpus. Product names, specifications, settings, site layout, screenshots and operational examples are synthetic. The two photographic illustrations were generated with AI. The floor plan, alignment diagram, architecture and console image were drawn programmatically; tables and body text were authored for this document.

No external photographs, customer information, credentials or confidential source documents were used. The document is intended to be included in public demonstration repositories and blog examples. Identify it as synthetic when presenting results; do not cite the invented specifications as real product claims.

### 12.3 Structure and version notes

Release 1.0 contains 12 numbered sections, six numbered figures and 12 numbered tables. Section headings, figure captions and table titles provide stable anchors. Front matter is separate from the product sections. Reading a section may still require a referenced table or figure from another section.